Building a User Interface

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Module Outline

Know the WinUI layout panels

- Build a user interface with the Grid

Use layout properties to position elements

Set attached properties in

- XAML
- C#

Know the WinUI Layout Panels

StackPanel

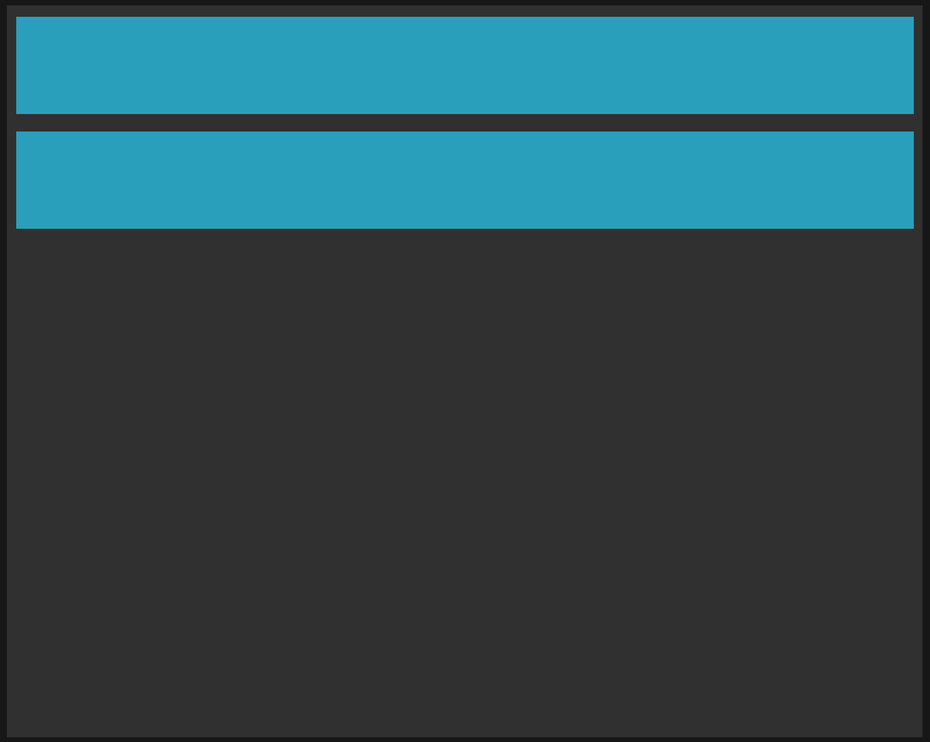
Grid

Canvas

Know the WinUI Layout Panels

StackPanel | Grid | Canvas

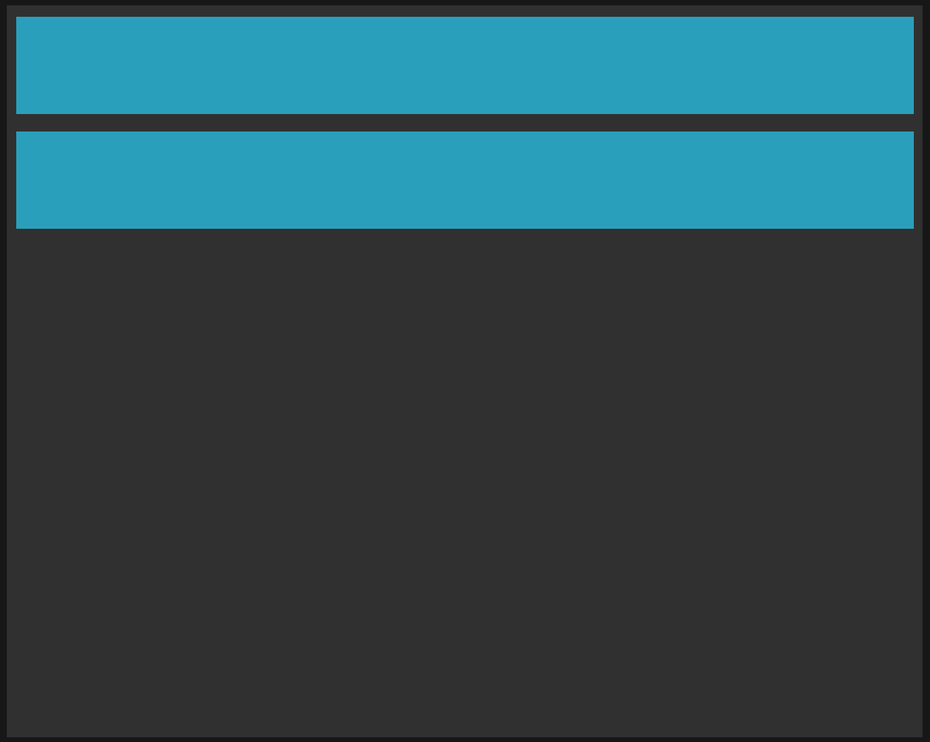
```
<StackPanel>
  <StackPanel.Children>
    <Rectangle Fill="LightBlue"
      Height="20" Margin="2"/>
    <Rectangle Fill="LightBlue"
      Height="20" Margin="2"/>
  </StackPanel.Children>
</StackPanel>
```



Know the WinUI Layout Panels

StackPanel | Grid | Canvas

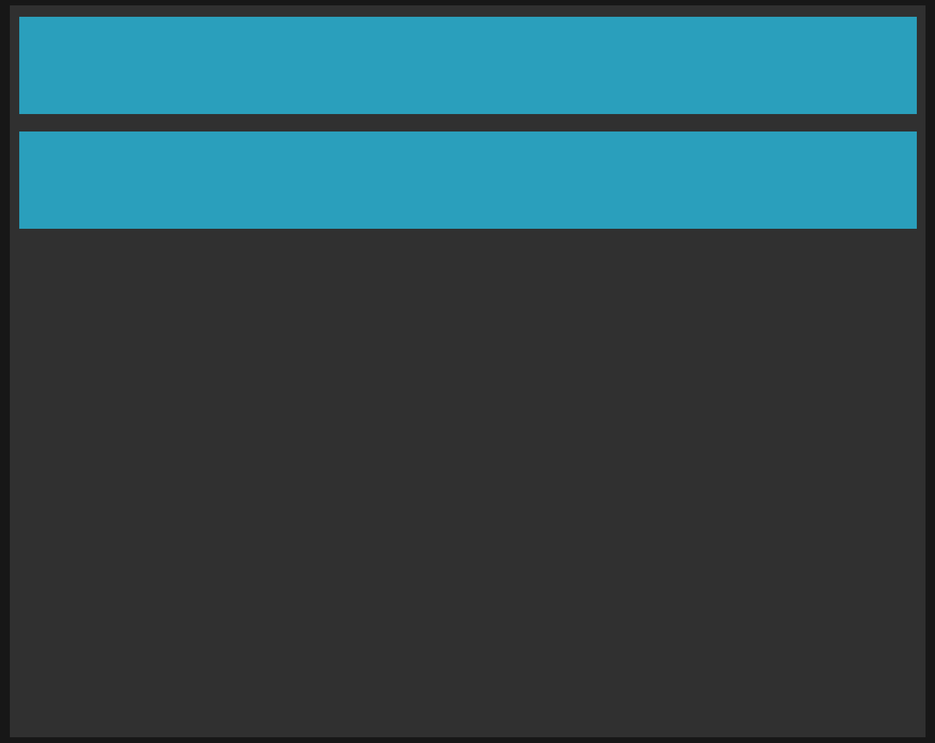
```
<StackPanel>
  <StackPanel.Children>
    <Rectangle Fill="LightBlue"
      Height="20" Margin="2"/>
    <Rectangle Fill="LightBlue"
      Height="20" Margin="2"/>
  </StackPanel.Children>
</StackPanel>
```



Know the WinUI Layout Panels

StackPanel | Grid | Canvas

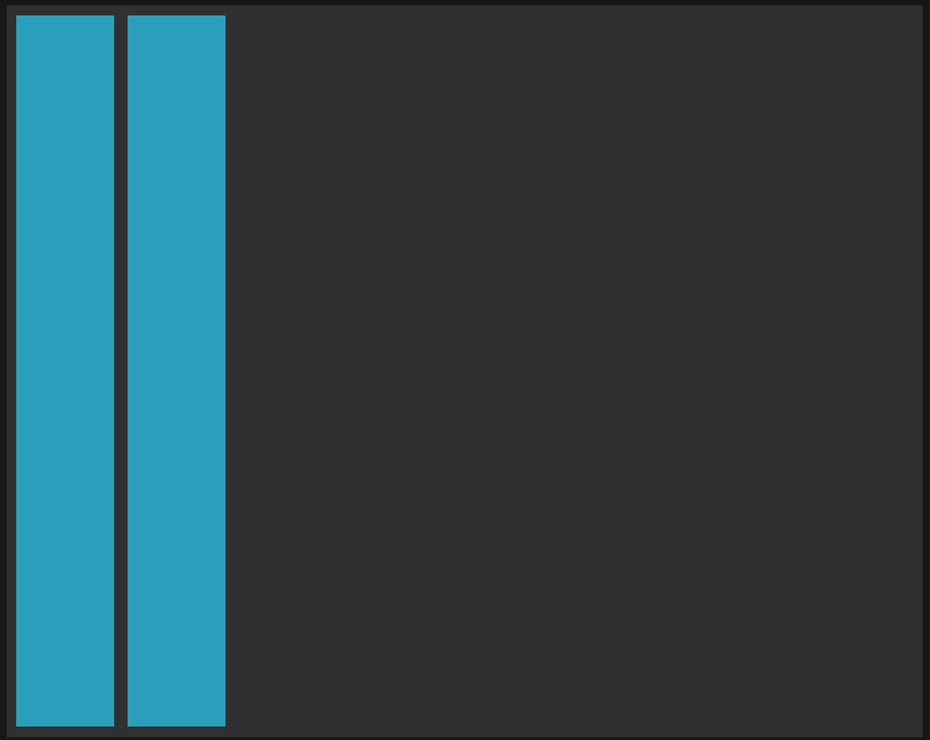
```
<StackPanel>
  <Rectangle Fill="LightBlue"
    Height="20" Margin="2"/>
  <Rectangle Fill="LightBlue"
    Height="20" Margin="2"/>
</StackPanel>
```



Know the WinUI Layout Panels

StackPanel | Grid | Canvas

```
<StackPanel Orientation="Horizontal">
  <Rectangle Fill="LightBlue"
    Width="20" Margin="2"/>
  <Rectangle Fill="LightBlue"
    Width="20" Margin="2"/>
</StackPanel>
```



Know the WinUI Layout Panels

StackPanel | **Grid** | Canvas

```
<Grid>
  <Grid.RowDefinitions>
    <RowDefinition/>
    <RowDefinition/>
  </Grid.RowDefinitions>
  <Grid.ColumnDefinitions>
    <ColumnDefinition/>
    <ColumnDefinition/>
  </Grid.ColumnDefinitions>


</Grid>
```

Know the WinUI Layout Panels

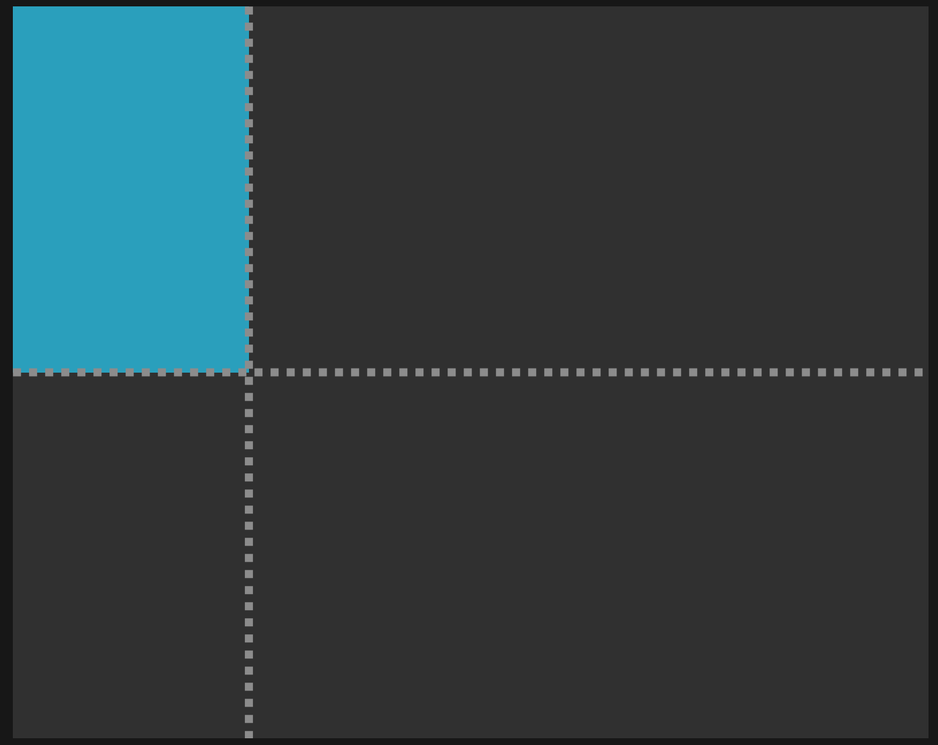
StackPanel | **Grid** | Canvas

```
<Grid>
  <Grid.RowDefinitions>
    <RowDefinition/>
    <RowDefinition/>
  </Grid.RowDefinitions>
  <Grid.ColumnDefinitions>
    <ColumnDefinition Width="40"/>
    <ColumnDefinition/>
  </Grid.ColumnDefinitions>


</Grid>
```

Know the WinUI Layout Panels

StackPanel | **Grid** | Canvas

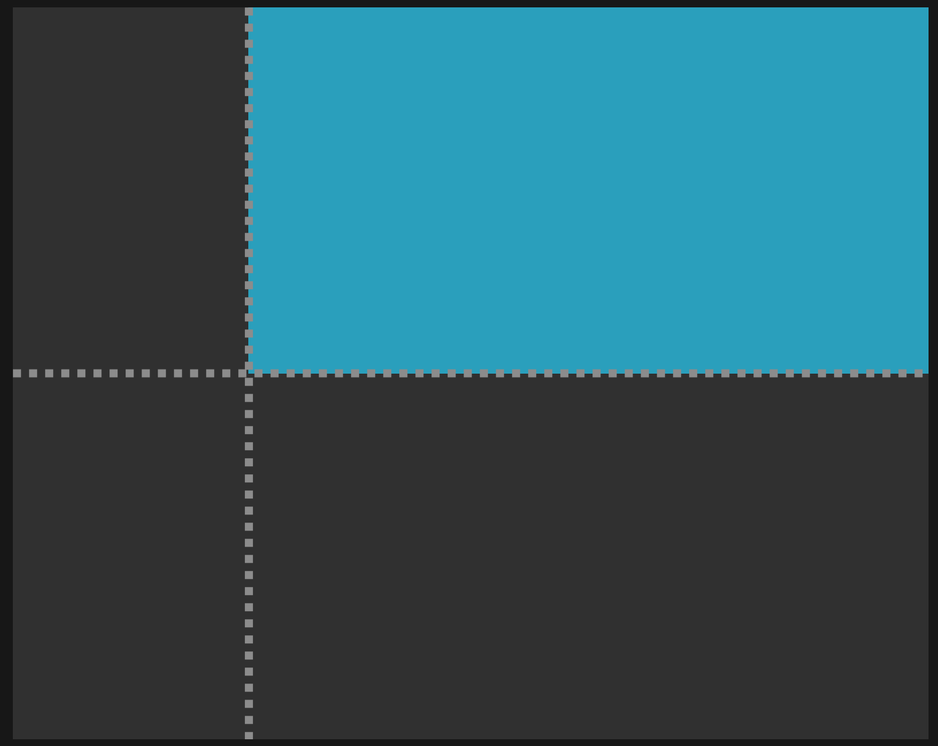
```
<Grid>
  <Grid.RowDefinitions>
    <RowDefinition/>
    <RowDefinition/>
  </Grid.RowDefinitions>
  <Grid.ColumnDefinitions>
    <ColumnDefinition Width="40"/>
    <ColumnDefinition/>
  </Grid.ColumnDefinitions>
  <Rectangle Fill="LightBlue"/>

</Grid>
```



Know the WinUI Layout Panels

StackPanel | **Grid** | Canvas

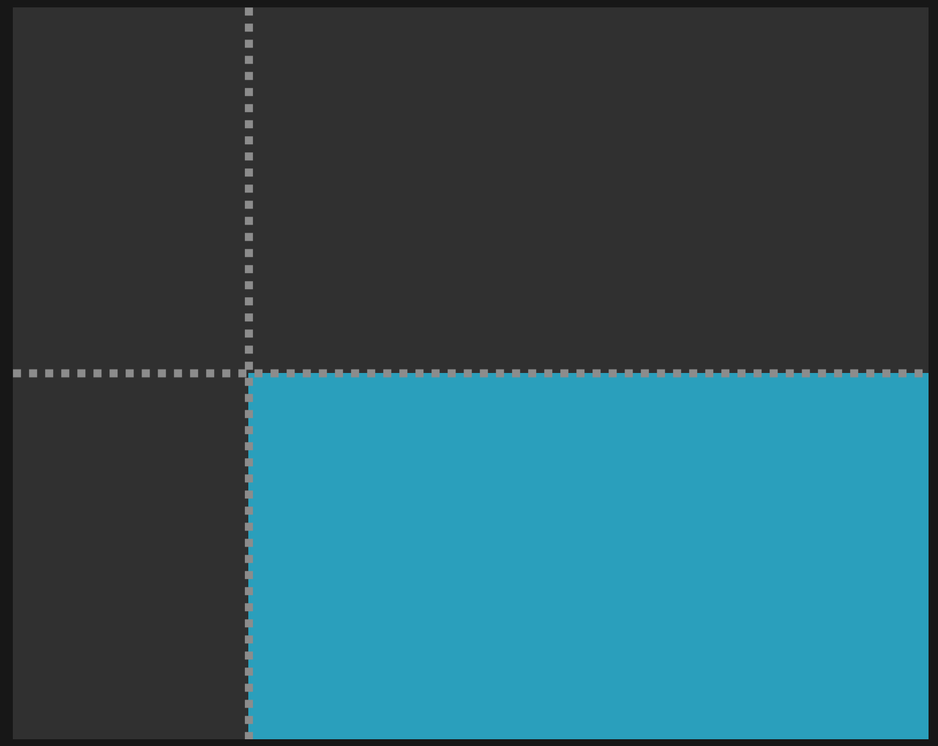
```
<Grid>
  <Grid.RowDefinitions>
    <RowDefinition/>
    <RowDefinition/>
  </Grid.RowDefinitions>
  <Grid.ColumnDefinitions>
    <ColumnDefinition Width="40"/>
    <ColumnDefinition/>
  </Grid.ColumnDefinitions>
  <Rectangle Fill="LightBlue"
    Grid.Column="1"/>
</Grid>
```



Know the WinUI Layout Panels

StackPanel | **Grid** | Canvas

```
<Grid>
  <Grid.RowDefinitions>
    <RowDefinition/>
    <RowDefinition/>
  </Grid.RowDefinitions>
  <Grid.ColumnDefinitions>
    <ColumnDefinition Width="40"/>
    <ColumnDefinition/>
  </Grid.ColumnDefinitions>
  <Rectangle Fill="LightBlue"
    Grid.Column="1" Grid.Row="1"/>
</Grid>
```



Know the WinUI Layout Panels

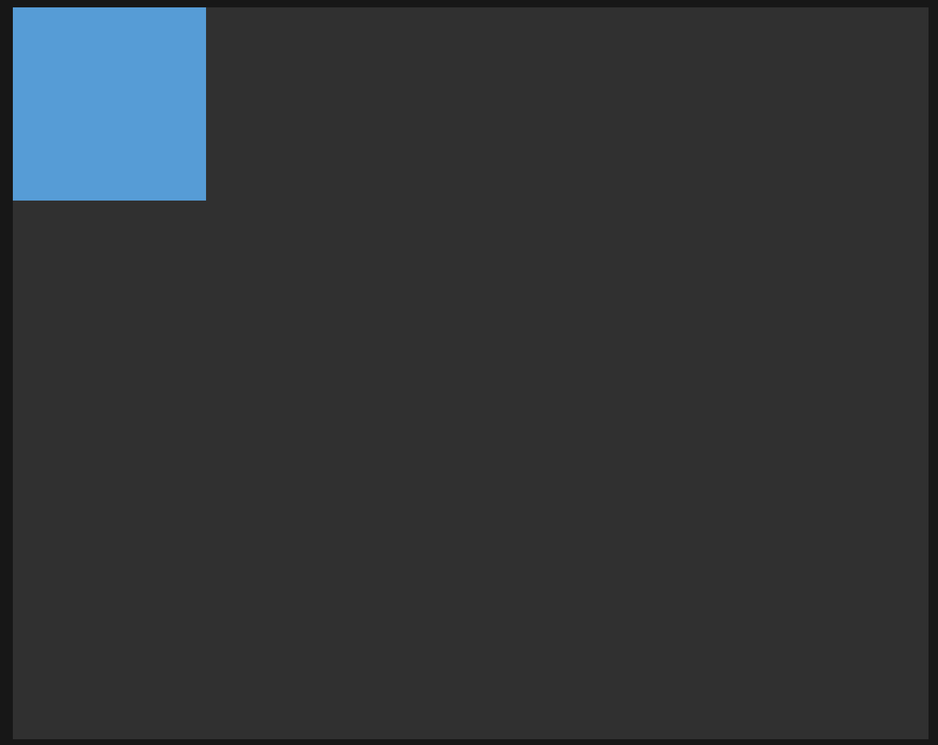
StackPanel | **Grid** | Canvas

```
<Grid>
  <Grid.RowDefinitions>
    <RowDefinition/>
    <RowDefinition/>
  </Grid.RowDefinitions>
  <Grid.ColumnDefinitions>
    <ColumnDefinition Width="40"/>
    <ColumnDefinition/>
  </Grid.ColumnDefinitions>
  <StackPanel Grid.Column="1">
  </StackPanel>
</Grid>
```

Know the WinUI Layout Panels

StackPanel | Grid | **Canvas**

```
<Canvas>
  <Rectangle Fill="LightBlue"
    Height="50" Width="50"/>



</Canvas>
```



Know the WinUI Layout Panels

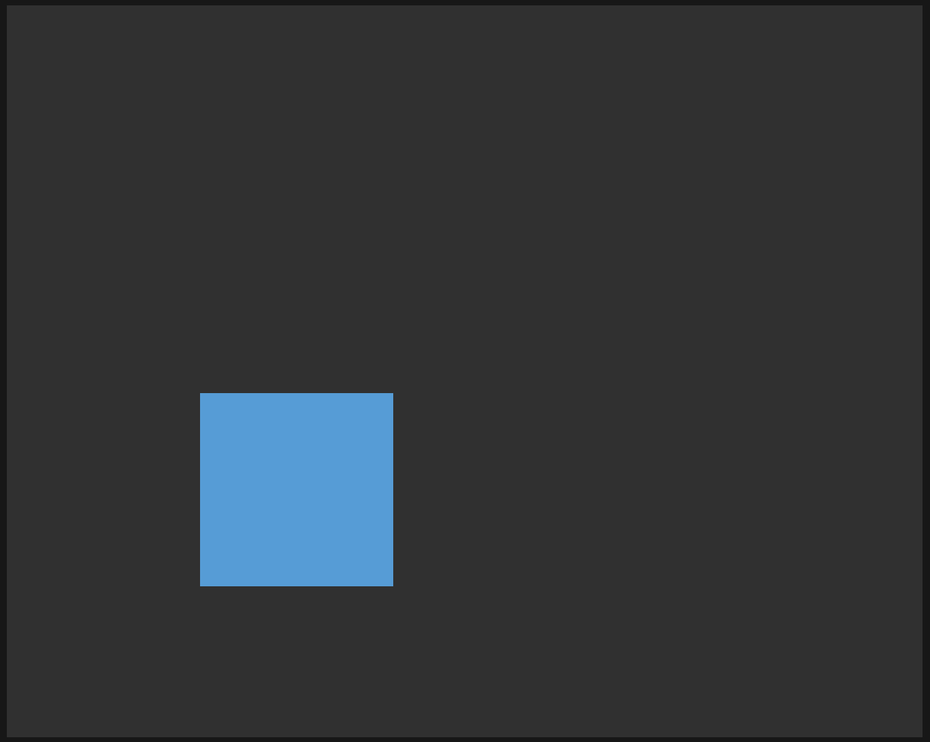
StackPanel | Grid | **Canvas**

```
<Canvas>
  <Rectangle Fill="LightBlue"
    Height="50" Width="50“
    Canvas.Left="50"/>



</Canvas>
```

Know the WinUI Layout Panels

StackPanel | Grid | **Canvas**

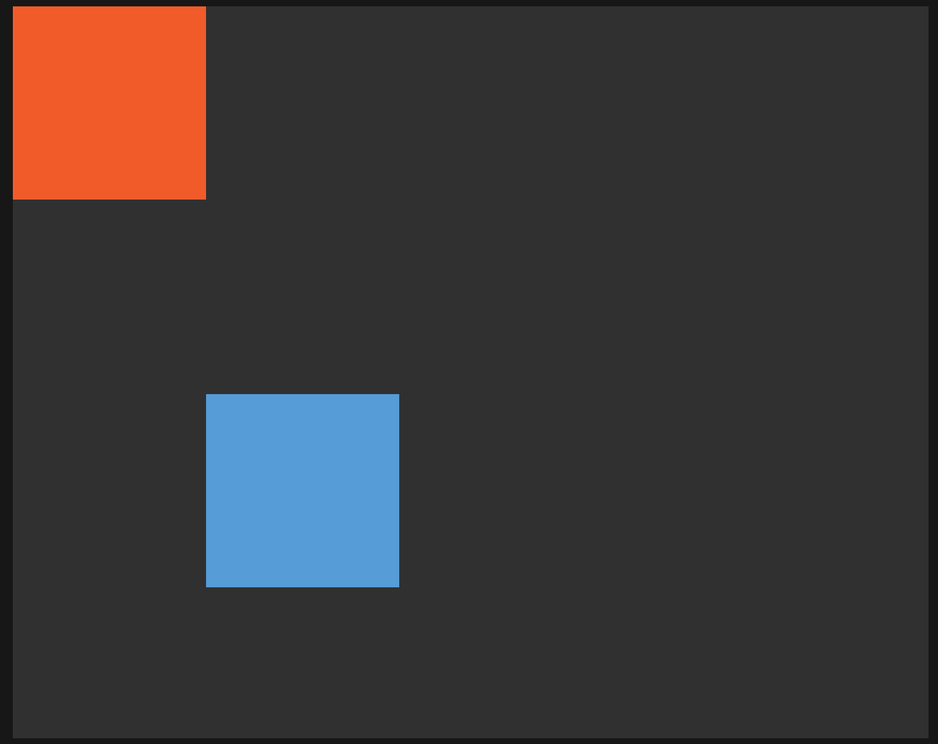
```
<Canvas>
  <Rectangle Fill="LightBlue"
    Height="50" Width="50“
    Canvas.Left="50"
    Canvas.Top="100"/>



</Canvas>
```



Know the WinUI Layout Panels

StackPanel | Grid | **Canvas**

```
<Canvas>
  <Rectangle Fill="LightBlue"
    Height="50" Width="50“
    Canvas.Left="50"
    Canvas.Top="100"/>
  <Rectangle Fill="Orange"
    Height="50" Width="50"/>


</Canvas>
```



Know the WinUI Layout Panels

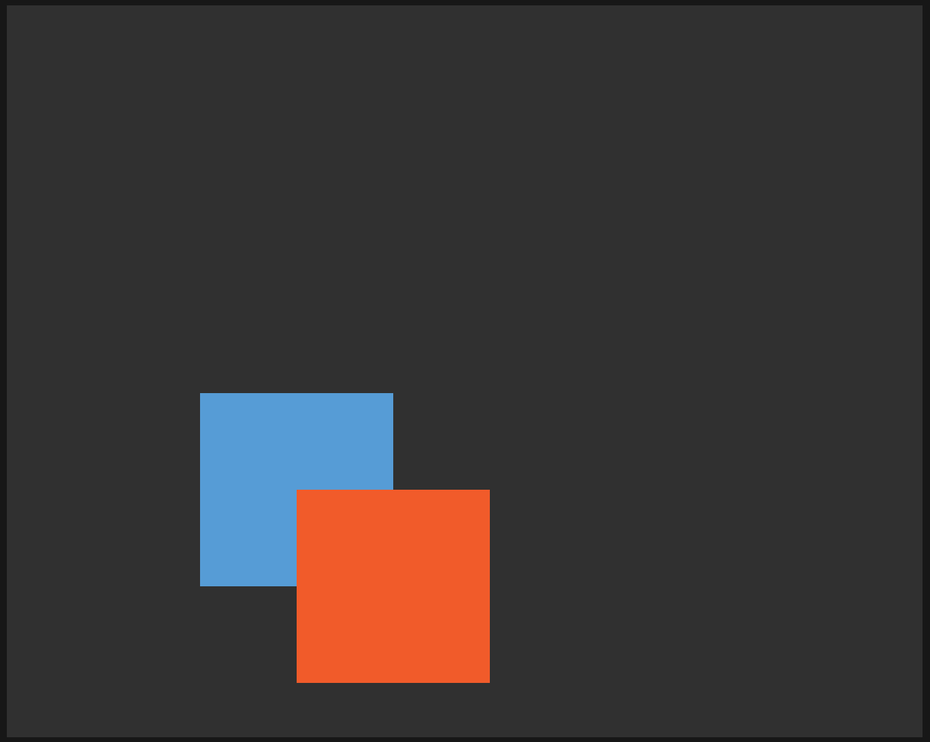
StackPanel | Grid | **Canvas**

```
<Canvas>
  <Rectangle Fill="LightBlue"
    Height="50" Width="50“
    Canvas.Left="50"
    Canvas.Top="100"/>
  <Rectangle Fill="Orange"
    Height="50" Width="50"
    Canvas.Left="75"/>

</Canvas>
```

Know the WinUI Layout Panels

StackPanel | Grid | **Canvas**

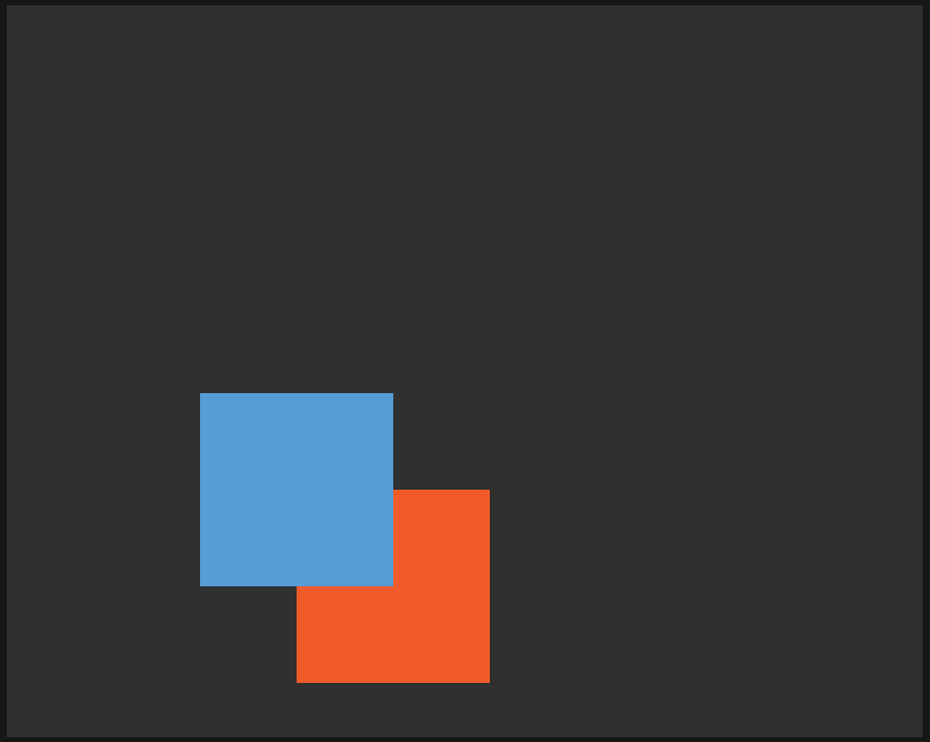
```
<Canvas>
  <Rectangle Fill="LightBlue"
    Height="50" Width="50“
    Canvas.Left="50"
    Canvas.Top="100"/>
  <Rectangle Fill="Orange"
    Height="50" Width="50"
    Canvas.Left="75"
    Canvas.Top="125"/>
</Canvas>
```



Know the WinUI Layout Panels

StackPanel | Grid | **Canvas**

```
<Canvas>
  <Rectangle Fill="LightBlue"
    Height="50" Width="50“
    Canvas.Left="50" Canvas.ZIndex="1"
    Canvas.Top="100"/>
  <Rectangle Fill="Orange"
    Height="50" Width="50"
    Canvas.Left="75"
    Canvas.Top="125"/>
</Canvas>
```



Demo

Build a layout with the Grid

Understand the Size of Rows and Columns

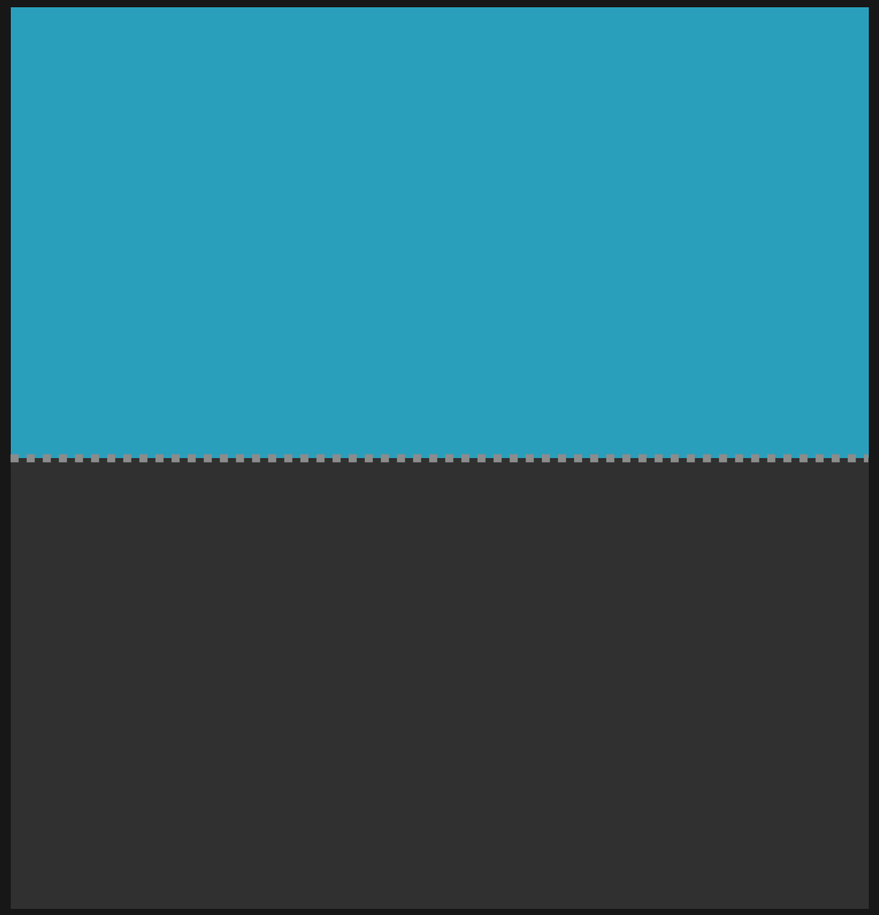
```
<Grid>
  <Grid.RowDefinitions>
    <RowDefinition/>
    <RowDefinition/>
  </Grid.RowDefinitions>

  <Rectangle Fill="LightBlue"/>

</Grid>
```

Understand the Size of Rows and Columns

```
<Grid>
  <Grid.RowDefinitions>
    <RowDefinition Height="*"/>
    <RowDefinition Height="*"/>
  </Grid.RowDefinitions>

  <Rectangle Fill="LightBlue"/>

</Grid>
```

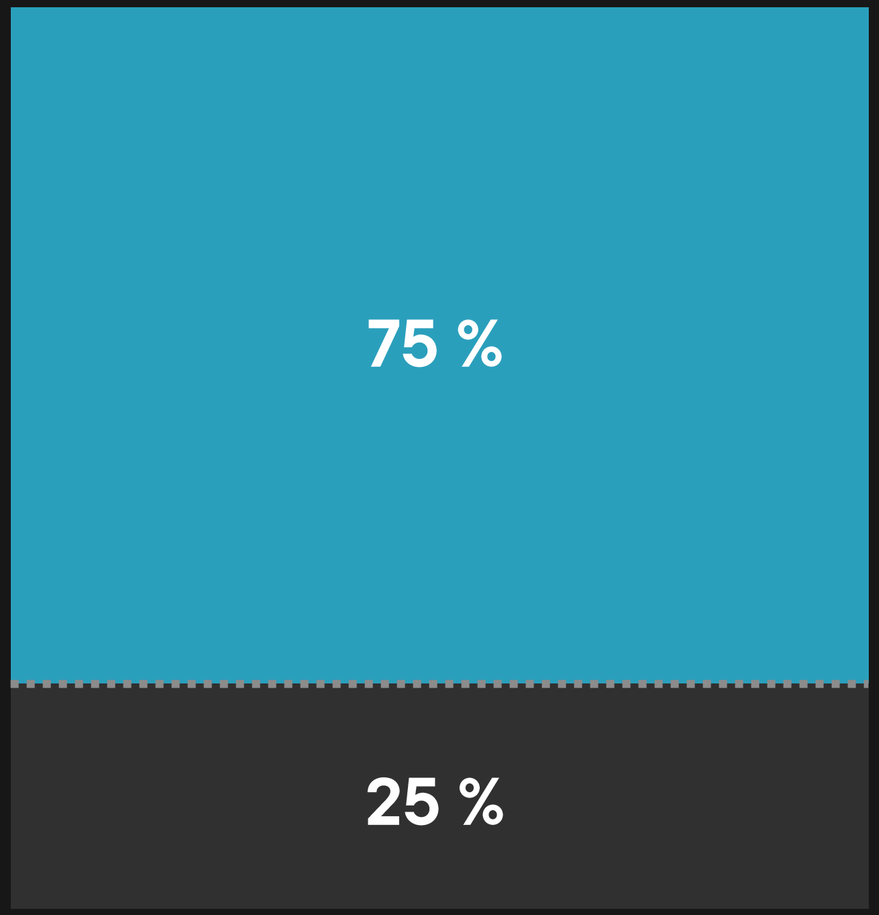
Understand the Size of Rows and Columns

```
<Grid>
  <Grid.RowDefinitions>
    <RowDefinition Height="*"/>
    <RowDefinition Height="*"/>
  </Grid.RowDefinitions>

  <Rectangle Fill="LightBlue"/>

</Grid>
```



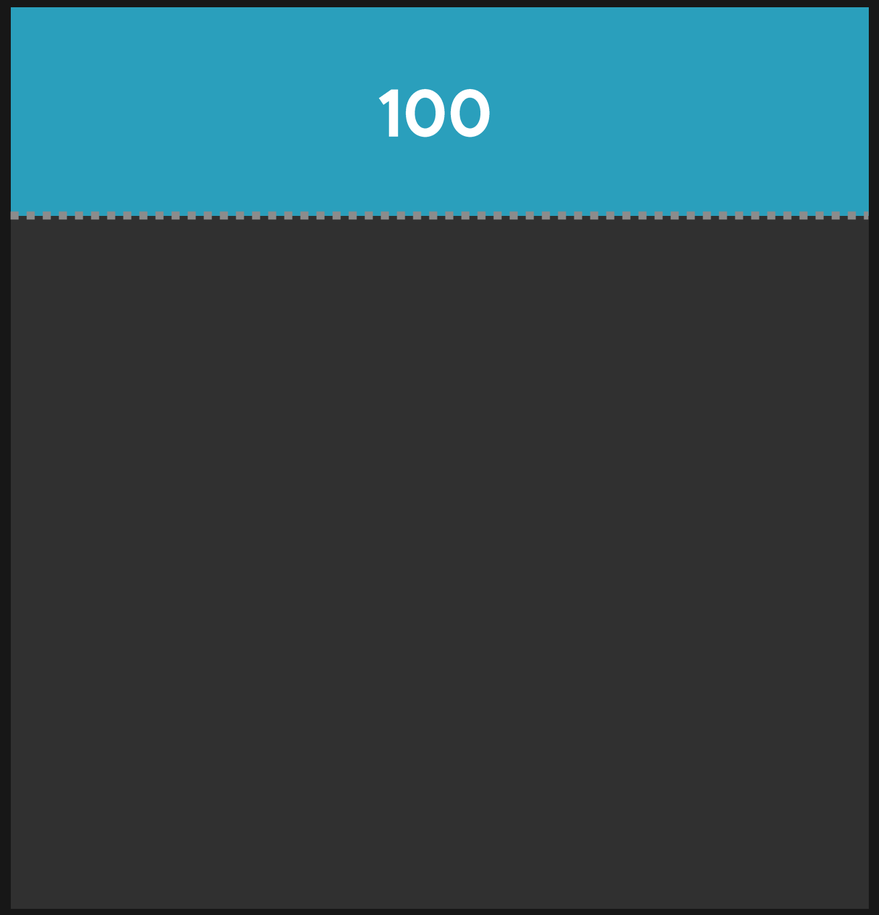
Understand the Size of Rows and Columns

```
<Grid>
  <Grid.RowDefinitions>
    <RowDefinition Height="3*"/>
    <RowDefinition Height="*"/>
  </Grid.RowDefinitions>

  <Rectangle Fill="LightBlue"/>

</Grid>
```



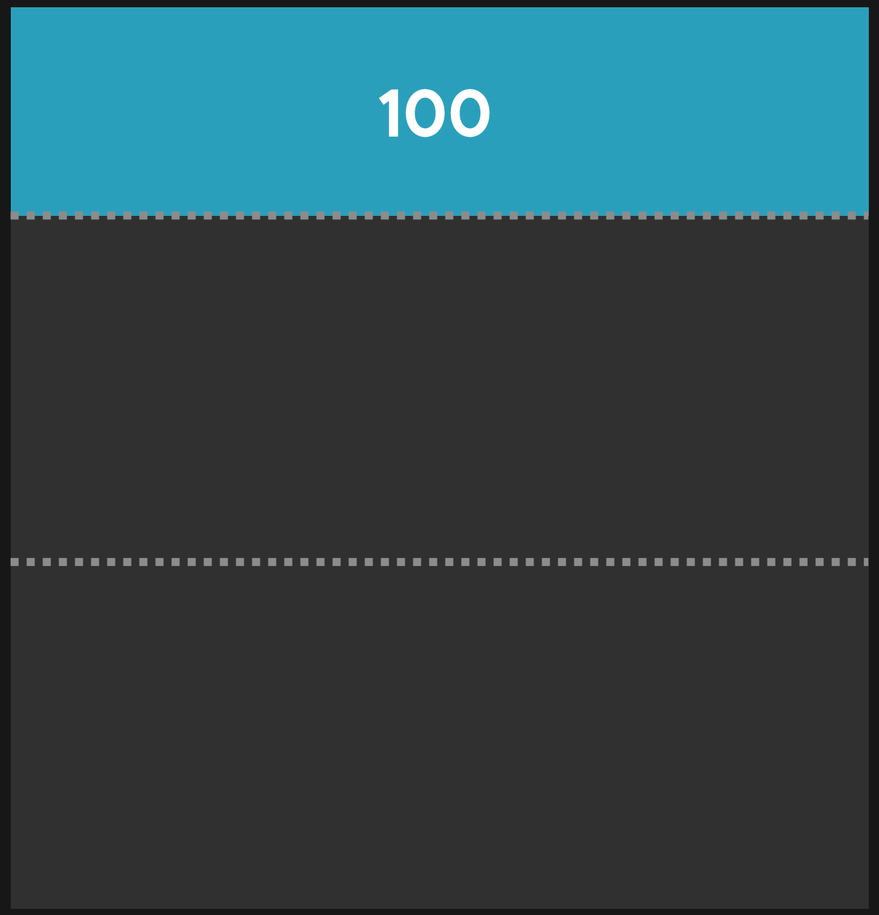
Understand the Size of Rows and Columns

```
<Grid>
  <Grid.RowDefinitions>
    <RowDefinition Height="100"/>
    <RowDefinition Height="*"/>
  </Grid.RowDefinitions>

  <Rectangle Fill="LightBlue"/>

</Grid>
```



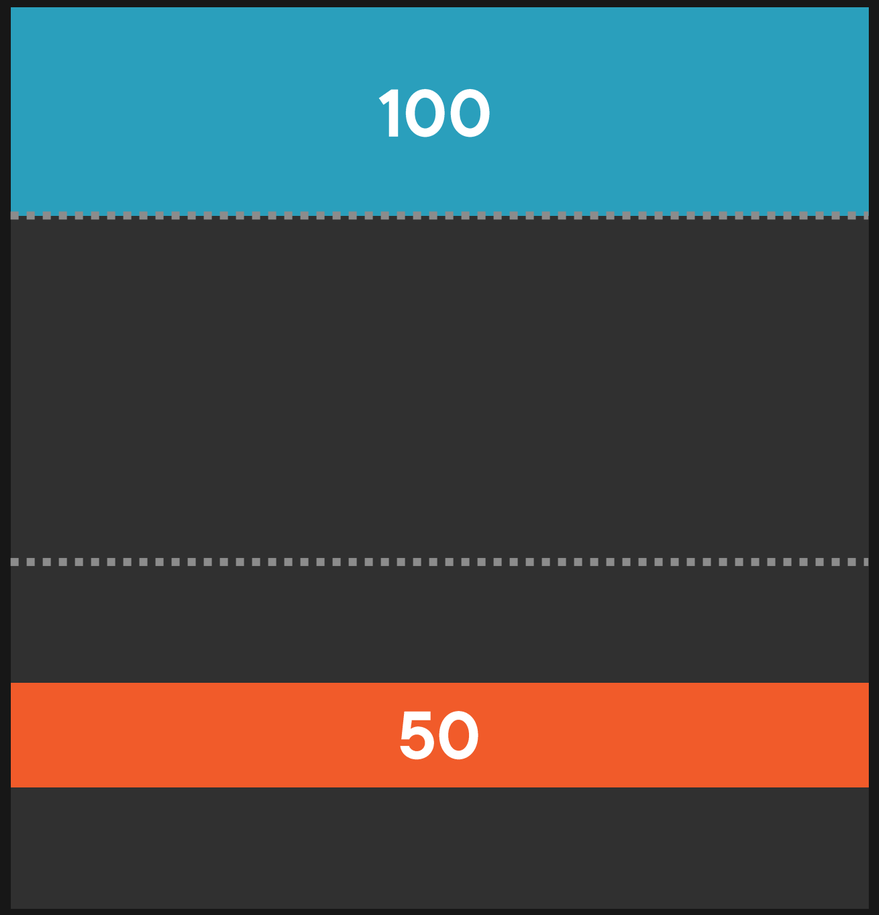
Understand the Size of Rows and Columns

```
<Grid>
  <Grid.RowDefinitions>
    <RowDefinition Height="100"/>
    <RowDefinition Height="*"/>
    <RowDefinition Height="*"/>
  </Grid.RowDefinitions>

  <Rectangle Fill="LightBlue"/>

</Grid>
```



Understand the Size of Rows and Columns

```
<Grid>
  <Grid.RowDefinitions>
    <RowDefinition Height="100"/>
    <RowDefinition Height="*"/>
    <RowDefinition Height="*"/>
  </Grid.RowDefinitions>

  <Rectangle Fill="LightBlue"/>

  <Rectangle Fill="Orange"
     Grid.Row="2" Height="50"/>

</Grid>
```

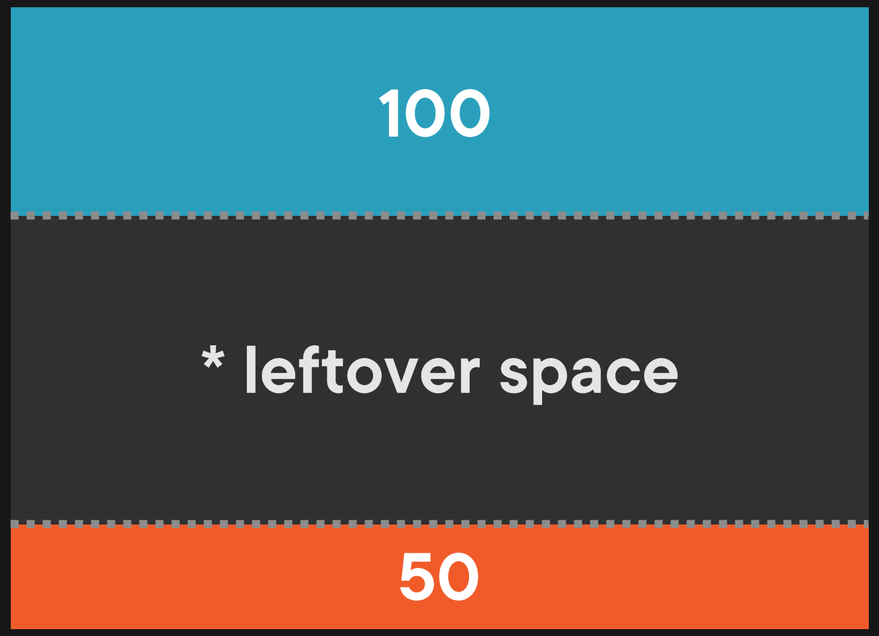


Understand the Size of Rows and Columns

```
<Grid>
  <Grid.RowDefinitions>
    <RowDefinition Height="100"/>
    <RowDefinition Height="*"/>
    <RowDefinition Height="Auto"/>
  </Grid.RowDefinitions>

  <Rectangle Fill="LightBlue"/>

  <Rectangle Fill="Orange"
     Grid.Row="2" Height="50"/>

</Grid>
```

Understand the Size of Rows and Columns

```
<Grid>
  <Grid.RowDefinitions>
    <RowDefinition Height="100"/>
    <RowDefinition Height="*"/>
    <RowDefinition Height="Auto"/>
  </Grid.RowDefinitions>

  <Rectangle Fill="LightBlue"/>

  <Rectangle Fill="Orange"
     Grid.Row="2" Height="50"/>

</Grid>
```



Understand the Size of Rows and Columns

```
<Grid>
  <Grid.RowDefinitions>
    <RowDefinition Height="100"/>
    <RowDefinition Height="*"/>
    <RowDefinition Height="Auto"/>
  </Grid.RowDefinitions>

  <Rectangle Fill="LightBlue"/>

  <Rectangle Fill="Orange"
     Grid.Row="2" Height="50"/>

</Grid>
```

Understand the Size of Rows and Columns

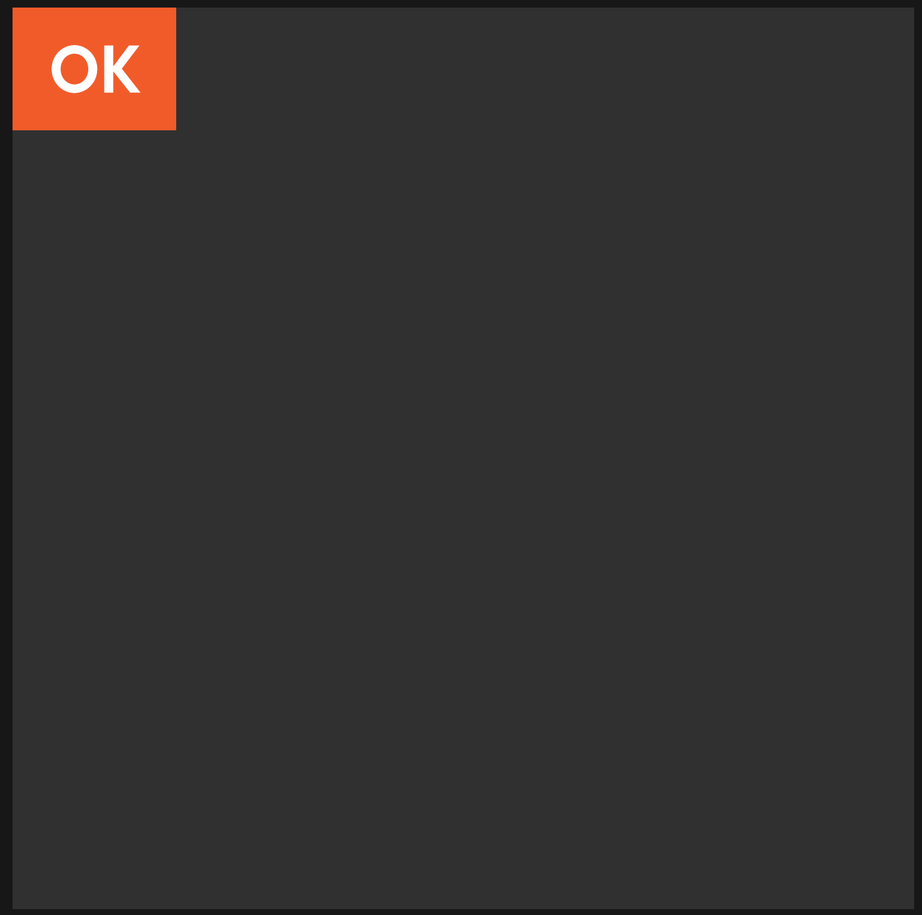
```
<Grid RowDefinitions="100,*,Auto">

  <Rectangle Fill="LightBlue"/>

  <Rectangle Fill="Orange"
     Grid.Row="2" Height="50"/>

</Grid>
```

Demo

Adjust the Grid in the application

Use Layout Properties to Position Elements

```
<Grid Width="200" Height="200">

  <Button Content="OK"
    HorizontalAlignment="Left"
    VerticalAlignment="Top"/>


</Grid>
```



Use Layout Properties to Position Elements

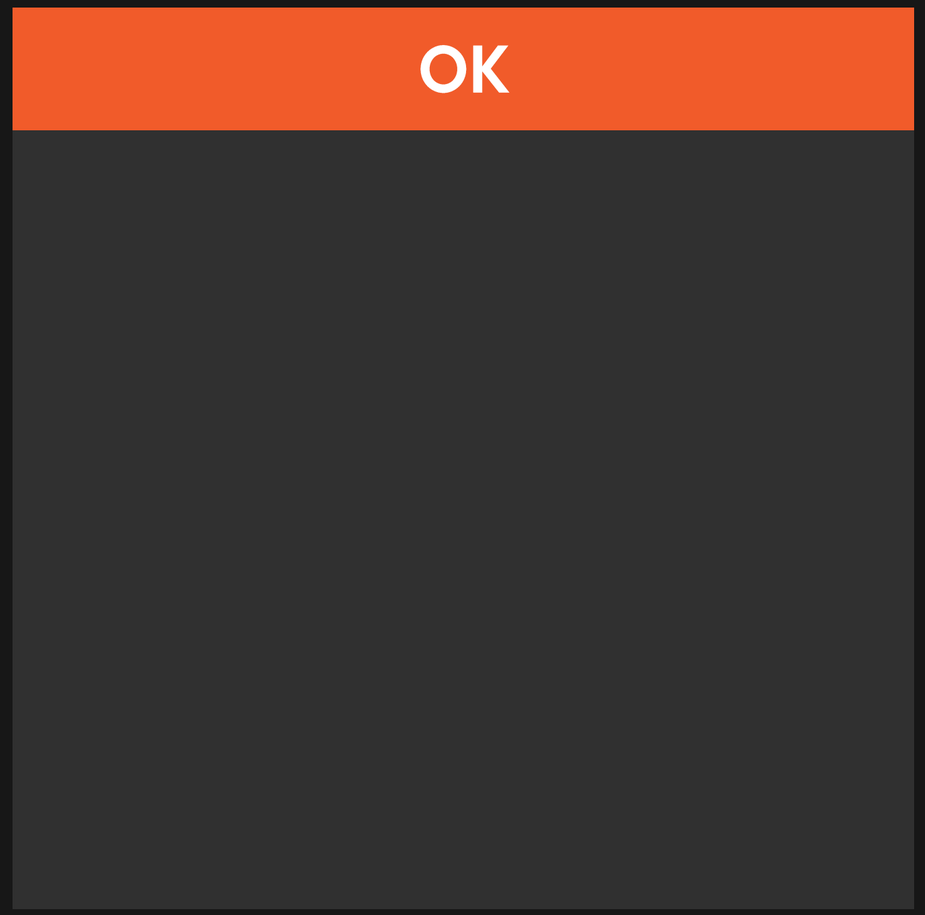
```
<Grid Width="200" Height="200">

  <Button Content="OK"
    HorizontalAlignment="Center"
    VerticalAlignment="Top"/>


</Grid>
```

Use Layout Properties to Position Elements

```
<Grid Width="200" Height="200">

  <Button Content="OK"
    HorizontalAlignment="Right"
    VerticalAlignment="Top"/>


</Grid>
```

Use Layout Properties to Position Elements

```
<Grid Width="200" Height="200">

  <Button Content="OK"
    HorizontalAlignment="Stretch"
    VerticalAlignment="Top"/>


</Grid>
```

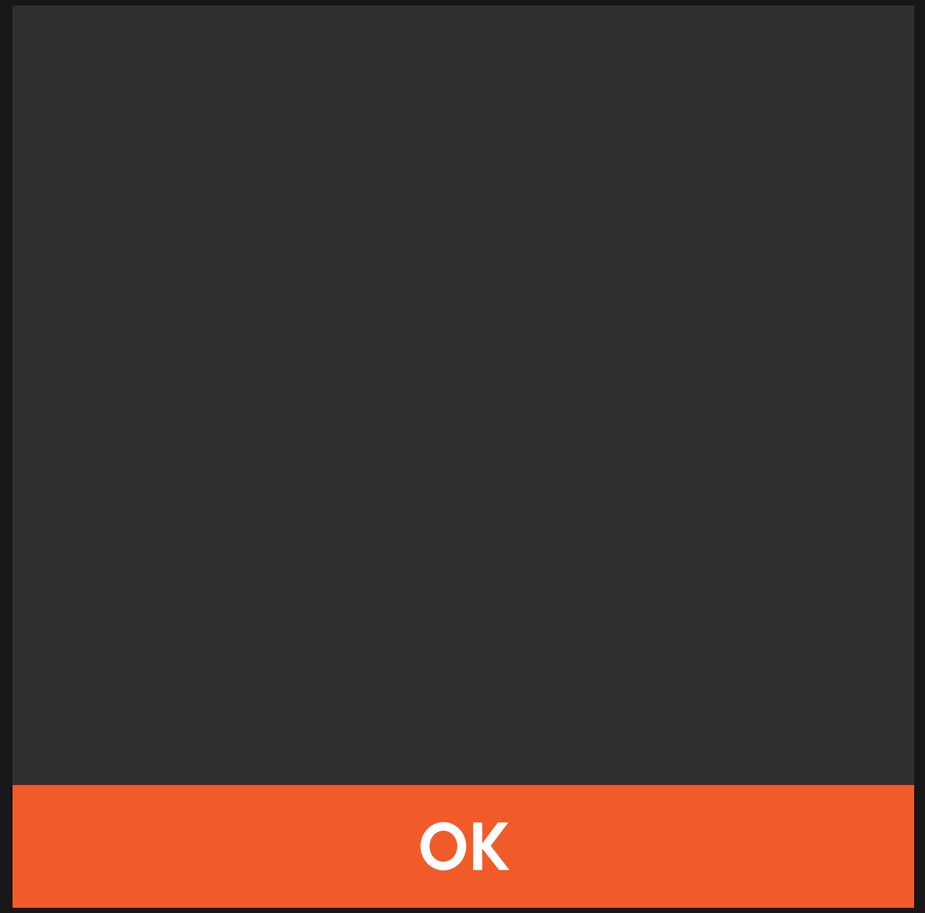


Use Layout Properties to Position Elements

```
<Grid Width="200" Height="200">

  <Button Content="OK"
    HorizontalAlignment="Stretch"
    VerticalAlignment="Center"/>


</Grid>
```

Use Layout Properties to Position Elements

```
<Grid Width="200" Height="200">

  <Button Content="OK"
    HorizontalAlignment="Stretch"
    VerticalAlignment="Bottom"/>


</Grid>
```



Use Layout Properties to Position Elements

```
<Grid Width="200" Height="200">

  <Button Content="OK"
    HorizontalAlignment="Stretch"
    VerticalAlignment="Stretch"/>


</Grid>
```

Use Layout Properties to Position Elements

```
<Grid Width="200" Height="200">

  <Button Content="OK"
    HorizontalAlignment="Stretch"
    VerticalAlignment="Stretch"
    Width="50"/>

</Grid>
```

Use Layout Properties to Position Elements

```
<Grid Width="200" Height="200">

  <Button Content="OK"
    HorizontalAlignment="Stretch"
    VerticalAlignment="Stretch"
    Width="50" Height="25"/>

</Grid>
```

Use Layout Properties to Position Elements

```
<Grid Width="200" Height="200">

  <Button Content="OK"
    HorizontalAlignment="Stretch"
    VerticalAlignment="Stretch"/>


</Grid>
```

Use Layout Properties to Position Elements

```
<Grid Width="200" Height="200">

  <Button Content="OK"
    HorizontalAlignment="Stretch"
    VerticalAlignment="Stretch"
    Margin="50"/>

</Grid>
```

Use Layout Properties to Position Elements

```
<Grid Width="200" Height="200">

  <Button Content="OK"
    HorizontalAlignment="Stretch"
    VerticalAlignment="Stretch"
    Margin="50 10"/>

</Grid>
```

Use Layout Properties to Position Elements

```
<Grid Width="200" Height="200">

  <Button Content="OK"
    HorizontalAlignment="Stretch"
    VerticalAlignment="Stretch"
    Margin="50 10 50 0"/>

</Grid>
```

Use Layout Properties to Position Elements

```
<Grid Width="200" Height="200">

  <Button Content="OK"
    HorizontalAlignment="Stretch"
    VerticalAlignment="Stretch"
    Margin="50,10,50,0"/>

</Grid>
```

Demo

Position elements in the header of the Customers application

Demo

Center the header

Demo

Create a nested Grid for the navigation

Demo

Ensure the ListView can scroll horizontally

Set Attached Properties in XAML

Attribute syntax

```
<Button Content="OK"/>
```

Property element syntax

```
<Button>
  <Button.Content>
    OK
  </Button.Content>
</Button>
```

Set Attached Properties in XAML

Attribute syntax

```
<Button Grid.Row="1"/>
```

Property element syntax

```
<Button>
  <Button.Content>
    OK
  </Button.Content>
</Button>
```

Set Attached Properties in XAML

Attribute syntax

```
<Button Grid.Row="1"/>
```

Property element syntax

```
<Button>
  <Grid.Row>
    1
  </Grid.Row>
</Button>
```

Set Attached Properties in C#

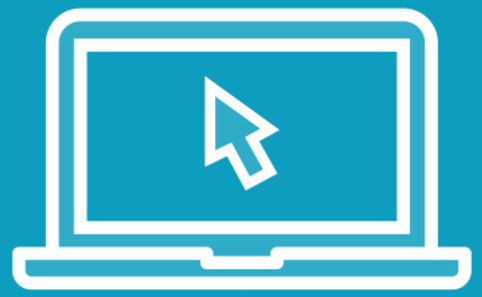
XAML

```
<Button Grid.Row="1"/>
```

C#

```
var btn = new Button();
btn.SetValue(Grid.RowProperty, 1);

var row =
  (int)btn.GetValue(Grid.RowProperty);
```

Demo



Set attached properties in C#

Move the navigation to the other side

Demo

Access attached properties with static methods

Summary

Use the WinUI layout panels

- Grid, StackPanel, Canvas

Position elements with layout properties

Set attached properties in

- XAML
- C#

Up Next:
Organizing Code with UserControls